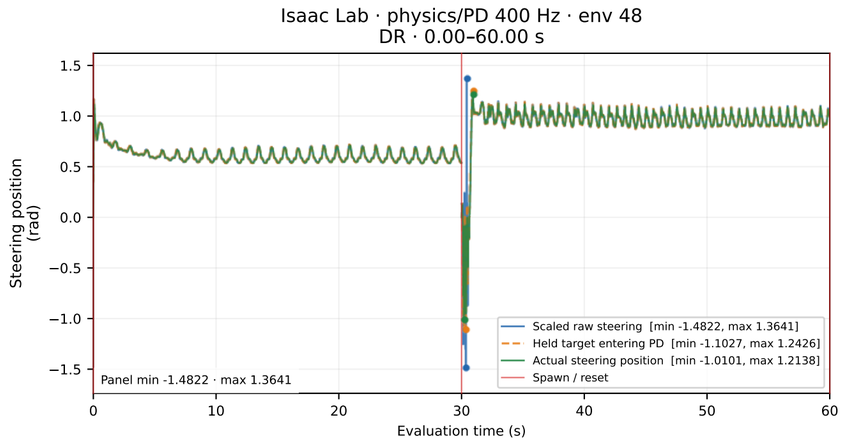
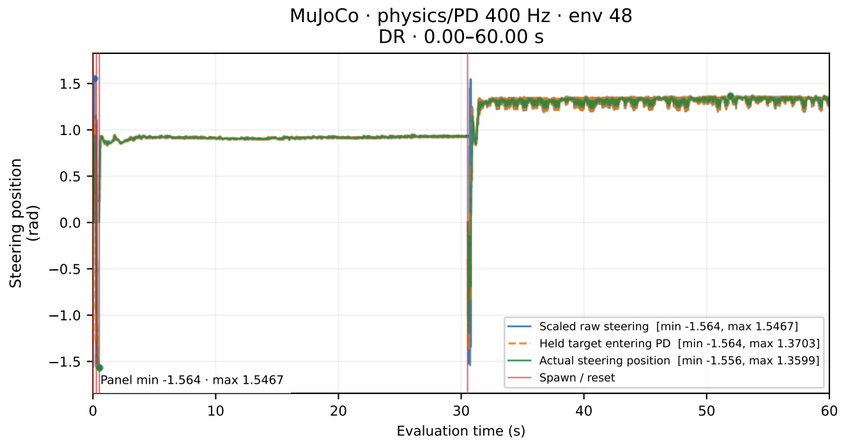
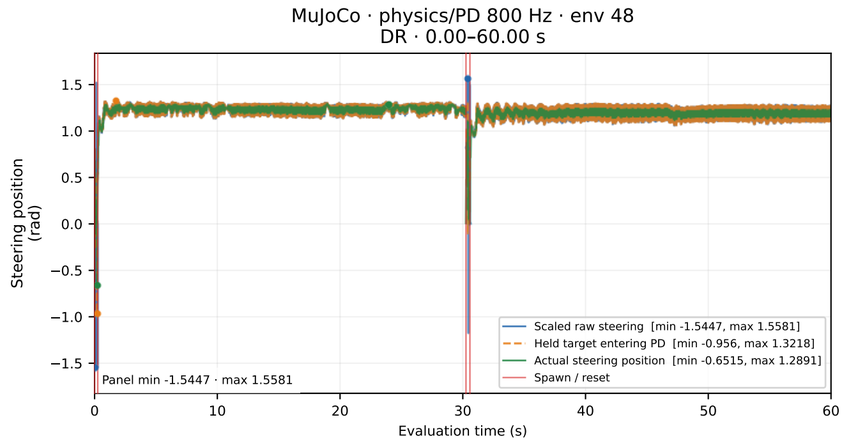
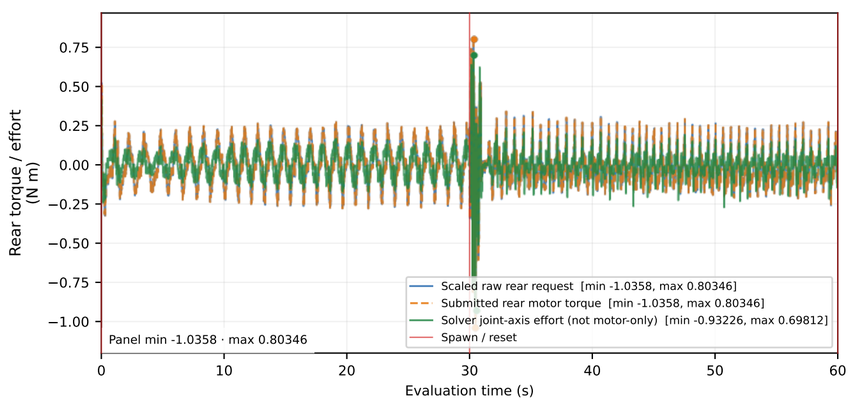
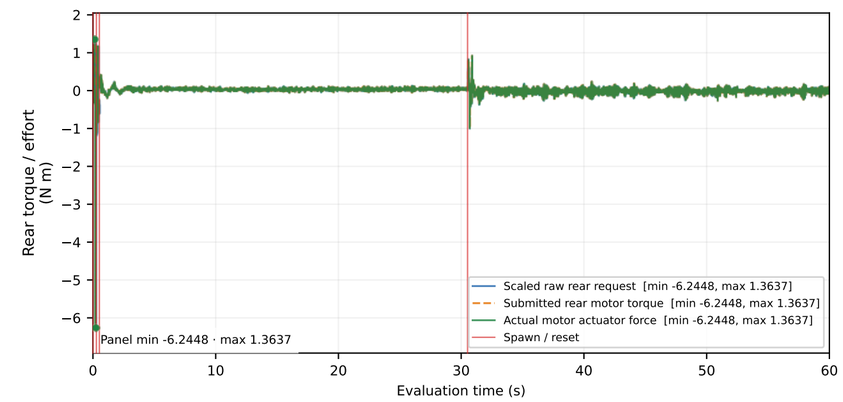
model_2500 · Demand versus actuation · env 48 · full 60 s

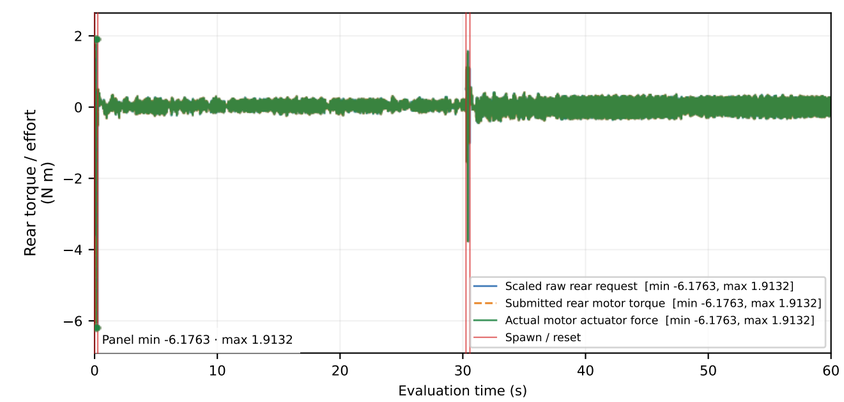












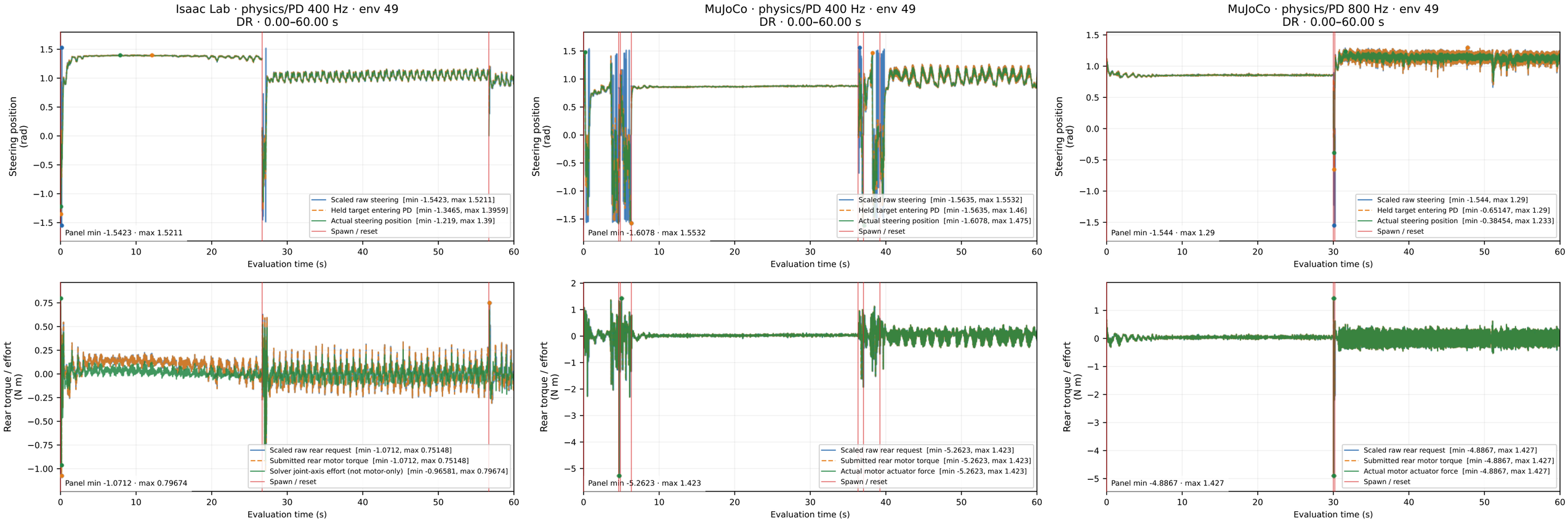
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 48; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.08753, 1.17826, 0.995849, 1.00857]
  CoM XYZ mm [-0.618026, 4.98509, 0.135079]
  mu_s [0.703711, 0.726175, 0.787792, 0.787792]
  mu_d [0.703711, 0.558574, 0.617134, 0.617134]
  Kp [steer,rear] [10.6269, 0]; Kd [0.208986, 0]
  spawn roll 0.0653718 rad; rate -0.159716 rad/s
  gyro bias XYZ rad/s [3.62393e-05, -9.55029e-05, -0.000273433]
  gravity bias YZ [-0.00302836, -0.00224948]
E2 [30.00, 60.00) s:
  mass x [1.14941, 1.15342, 1.11081, 0.896198]
  CoM XYZ mm [-2.72126, 1.15197, 1.6517]
  mu_s [0.579974, 0.693205, 0.547743, 0.693205]
  mu_d [0.579974, 0.693205, 0.547743, 0.693205]
  Kp [steer,rear] [10.6851, 0]; Kd [0.18368, 0]
  spawn roll -0.0243969 rad; rate 0.0606264 rad/s
  gyro bias XYZ rad/s [-0.000184089, -0.00030335, -0.000277833]
  gravity bias YZ [0.00169077, -0.00102462]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 48; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [0.948933, 1.07276, 1.00741, 0.96734]
  CoM XYZ mm [1.96312, -3.74609, 1.99845]
  mu_s [0.559972, 0.71155, 0.669571, 0.640016]
  mu_d [0.502209, 0.71155, 0.669571, 0.513141]
  Kp [steer,rear] [9.00075, 0]; Kd [0.213496, 0]
  spawn roll 0.0828675 rad; rate -0.189681 rad/s
  gyro bias XYZ rad/s [0.000127252, 0.000128662, -0.000120542]
  gravity bias YZ [0.00266815, -0.00159674]
E2 [0.28, 0.52) s:
  mass x [0.969414, 1.10477, 0.87968, 0.969751]
  CoM XYZ mm [-3.70687, 1.66721, -0.934434]
  mu_s [0.568073, 0.723429, 0.640867, 0.704749]
  mu_d [0.568073, 0.723429, 0.556841, 0.541926]
  Kp [steer,rear] [9.89196, 0]; Kd [0.213456, 0]
  spawn roll -0.0883098 rad; rate -0.0289943 rad/s
  gyro bias XYZ rad/s [6.7487e-05, -5.02594e-05, 0.000127656]
  gravity bias YZ [0.00475094, 0.0020392]
E3 [0.52, 30.52) s:
  mass x [1.10132, 0.988294, 1.07585, 1.10371]
  CoM XYZ mm [2.41057, 2.85973, 0.25693]
  mu_s [0.640016, 0.757579, 0.704749, 0.541939]
  mu_d [0.513141, 0.70921, 0.541926, 0.534359]
  Kp [steer,rear] [10.7175, 0]; Kd [0.205582, 0]
  spawn roll 0.0529119 rad; rate -0.148663 rad/s
  gyro bias XYZ rad/s [0.000160925, 0.000172416, 0.000212644]
  gravity bias YZ [0.000404312, -0.00192323]
E4 [30.52, 60.00) s:
  mass x [0.952696, 0.990547, 0.948737, 0.931427]
  CoM XYZ mm [-4.97849, -3.83979, -0.657533]
  mu_s [0.559972, 0.633024, 0.597748, 0.640016]
  mu_d [0.502209, 0.568172, 0.597748, 0.513141]
  Kp [steer,rear] [9.14485, 0]; Kd [0.191679, 0]
  spawn roll -0.0652092 rad; rate -0.0960493 rad/s
  gyro bias XYZ rad/s [0.000207544, 0.00013045, -0.000430529]
  gravity bias YZ [0.000374311, 0.000344802]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 48; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.26) s:
  mass x [0.948933, 1.07276, 1.00741, 0.96734]
  CoM XYZ mm [1.96312, -3.74609, 1.99845]
  mu_s [0.559972, 0.71155, 0.669571, 0.640016]
  mu_d [0.502209, 0.71155, 0.669571, 0.513141]
  Kp [steer,rear] [9.00075, 0]; Kd [0.213496, 0]
  spawn roll 0.0828675 rad; rate -0.189681 rad/s
  gyro bias XYZ rad/s [0.000127252, 0.000128662, -0.000120542]
  gravity bias YZ [0.00266815, -0.00159674]
E2 [0.26, 30.26) s:
  mass x [0.808865, 1.01398, 0.994135, 1.00275]
  CoM XYZ mm [-4.31078, -1.99309, 2.79664]
  mu_s [0.640867, 0.732187, 0.559972, 0.597748]
  mu_d [0.556841, 0.631664, 0.502209, 0.597748]
  Kp [steer,rear] [10.9784, 0]; Kd [0.196941, 0]
  spawn roll -0.00321518 rad; rate 0.139784 rad/s
  gyro bias XYZ rad/s [0.000153575, -0.000135664, 1.55938e-05]
  gravity bias YZ [-0.00113177, 0.000514633]
E3 [30.26, 30.58) s:
  mass x [1.08386, 0.853643, 1.16262, 0.861685]
  CoM XYZ mm [2.43457, -3.03032, 2.42355]
  mu_s [0.541939, 0.71155, 0.733515, 0.749678]
  mu_d [0.534359, 0.71155, 0.558392, 0.741429]
  Kp [steer,rear] [10.4606, 0]; Kd [0.216125, 0]
  spawn roll 0.0840548 rad; rate 0.0711848 rad/s
  gyro bias XYZ rad/s [-8.50104e-05, -0.000309878, -9.46143e-05]
  gravity bias YZ [-0.0036056, -0.000550167]
E4 [30.58, 60.00) s:
  mass x [0.9917, 1.00228, 1.00319, 0.839121]
  CoM XYZ mm [0.515675, 0.182563, -1.45984]
  mu_s [0.631146, 0.640016, 0.640867, 0.749678]
  mu_d [0.631146, 0.513141, 0.556841, 0.741429]
  Kp [steer,rear] [9.1803, 0]; Kd [0.206165, 0]
  spawn roll 0.0543008 rad; rate 0.124664 rad/s
  gyro bias XYZ rad/s [-0.00024219, -0.000192722, -1.61808e-05]
  gravity bias YZ [0.00465794, 0.0006785]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 49 · full 60 s



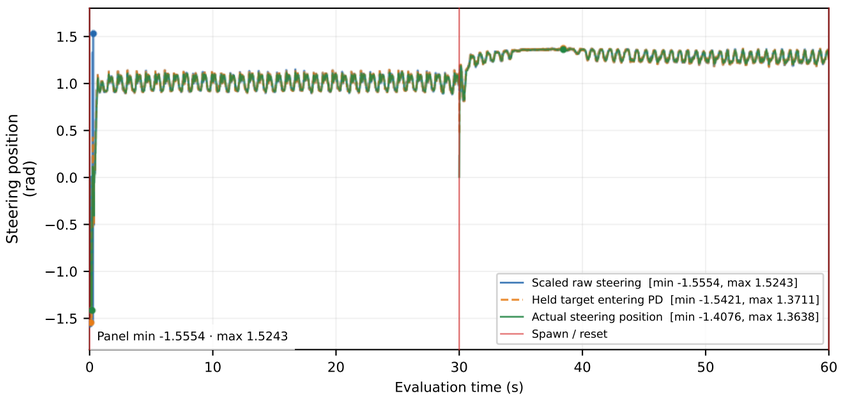
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

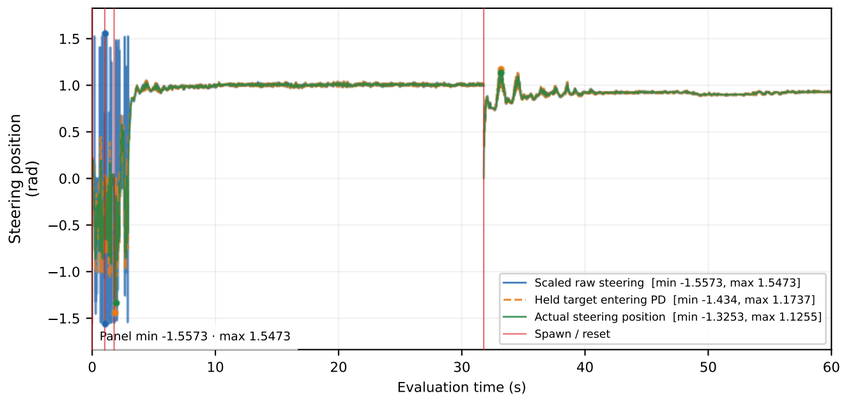
```
SAMPLED DR: env 49; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 26.68) s:
  mass x [1.16767, 1.02079, 1.07897, 0.814629]
  CoM XYZ mm [-3.47495, -3.38237, -3.22597]
  mu_s [0.547743, 0.598343, 0.774409, 0.760821]
  mu_d [0.547743, 0.598343, 0.561095, 0.670315]
  Kp [steer,rear] [10.5091, 0]; Kd [0.190662, 0]
  spawn roll -0.0669806 rad; rate -0.085151 rad/s
  gyro bias XYZ rad/s [0.000105286, 0.000217856, -4.12894e-06]
  gravity bias YZ [-0.000672248, 0.00182319]
E2 [26.68, 56.68) s:
  mass x [0.961654, 1.16869, 0.915261, 0.876968]
  CoM XYZ mm [3.00056, 0.714, 1.20407]
  mu_s [0.741039, 0.681711, 0.560535, 0.713129]
  mu_d [0.614405, 0.611756, 0.560535, 0.713129]
  Kp [steer,rear] [10.4558, 0]; Kd [0.180785, 0]
  spawn roll -0.0518287 rad; rate 0.136825 rad/s
  gyro bias XYZ rad/s [-0.000321651, -0.000150021, 0.000280684]
  gravity bias YZ [-0.00145164, 0.00206375]
E3 [56.68, 60.00) s:
  mass x [1.08793, 0.868082, 1.03885, 0.850176]
  CoM XYZ mm [-3.32075, 1.74008, 2.60136]
  mu_s [0.741039, 0.673378, 0.592046, 0.787792]
  mu_d [0.614405, 0.673378, 0.534947, 0.617134]
  Kp [steer,rear] [10.278, 0]; Kd [0.207587, 0]
  spawn roll 0.0610619 rad; rate -0.115938 rad/s
  gyro bias XYZ rad/s [-8.96865e-06, 0.00025883, 0.000231798]
  gravity bias YZ [0.00299946, -0.00336261]
Six significant digits shown; full precision: sampled_dr_values.json.
```

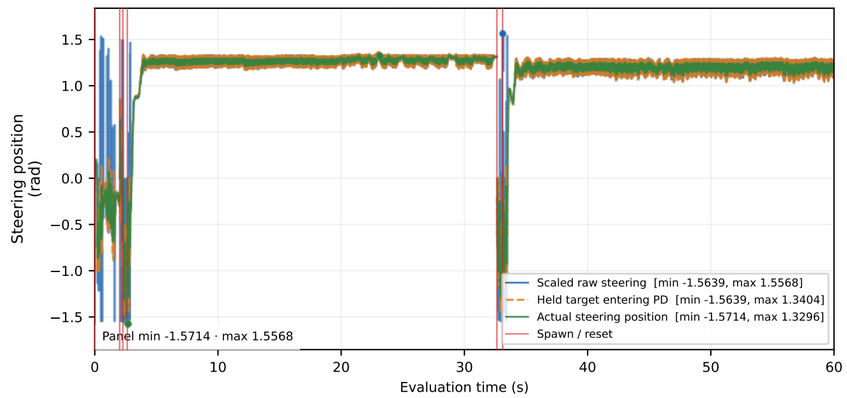
```
SAMPLED DR: env 49; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 4.64) s:
  mass x [0.913233, 1.1641, 1.12773, 0.91701]
  CoM XYZ mm [3.34854, 4.04615, -2.15763]
  mu_s [0.736077, 0.637675, 0.640016, 0.733515]
  mu_d [0.699455, 0.637675, 0.513141, 0.558392]
  Kp [steer,rear] [10.1476, 0]; Kd [0.213448, 0]
  spawn roll 0.0863189 rad; rate -0.0570891 rad/s
  gyro bias XYZ rad/s [0.000281771, -0.000166747, 0.000119675]
  gravity bias YZ [0.00174457, 0.00369669]
E2 [4.64, 4.86) s:
  mass x [1.19235, 1.08213, 1.02801, 1.05424]
  CoM XYZ mm [-1.72457, -4.12029, 2.35562]
  mu_s [0.541939, 0.546287, 0.736077, 0.637675]
  mu_d [0.534359, 0.546287, 0.699455, 0.637675]
  Kp [steer,rear] [9.4624, 0]; Kd [0.21421, 0]
  spawn roll 0.079729 rad; rate 0.119137 rad/s
  gyro bias XYZ rad/s [-0.000233366, -3.99513e-05, 0.000153254]
  gravity bias YZ [-0.00349055, 0.00278417]
E3 [4.86, 6.32) s:
  mass x [1.00931, 0.890937, 1.10744, 0.836926]
  CoM XYZ mm [4.30324, 4.22095, -2.89644]
  mu_s [0.568073, 0.704749, 0.637675, 0.616244]
  mu_d [0.568073, 0.541926, 0.637675, 0.586498]
  Kp [steer,rear] [10.8241, 0]; Kd [0.187122, 0]
  spawn roll 0.0697917 rad; rate 0.0812057 rad/s
  gyro bias XYZ rad/s [0.000127583, 5.33777e-05, -0.000312492]
  gravity bias YZ [0.000882079, 0.000244129]
E4 [6.32, 36.32) s:
  mass x [0.926608, 1.13568, 1.17766, 0.93652]
  CoM XYZ mm [2.38566, 3.78858, 2.70094]
  mu_s [0.71155, 0.631146, 0.733515, 0.732187]
  mu_d [0.71155, 0.631146, 0.558392, 0.631664]
  Kp [steer,rear] [9.62313, 0]; Kd [0.188571, 0]
  spawn roll 0.0609389 rad; rate 0.225905 rad/s
  gyro bias XYZ rad/s [0.000281571, -1.67543e-05, -8.8375e-05]
  gravity bias YZ [8.35985e-05, 0.00145331]
E5 [36.32, 37.04) s:
  mass x [0.839765, 1.11111, 0.801269, 0.816831]
  CoM XYZ mm [-4.50648, -3.85568, 3.59316]
  mu_s [0.559972, 0.749678, 0.757579, 0.640867]
  mu_d [0.502209, 0.741429, 0.70921, 0.556841]
  Kp [steer,rear] [9.09879, 0]; Kd [0.203822, 0]
  spawn roll 0.0164155 rad; rate -0.126637 rad/s
  gyro bias XYZ rad/s [-0.000320956, 6.99507e-05, -0.000274306]
  gravity bias YZ [0.00341725, 0.002529]
E6 [37.04, 39.22) s:
  mass x [1.03434, 1.08937, 1.01501, 0.951878]
  CoM XYZ mm [2.04634, -0.712359, -2.41383]
  mu_s [0.733515, 0.749678, 0.538976, 0.728342]
  mu_d [0.558392, 0.741429, 0.538976, 0.728342]
  Kp [steer,rear] [9.93877, 0]; Kd [0.208141, 0]
  spawn roll 0.0784524 rad; rate 0.166272 rad/s
  gyro bias XYZ rad/s [-0.000227385, -1.69427e-05, -0.000156127]
  gravity bias YZ [-0.000363731, -0.00107437]
E7 [39.22, 60.00) s:
  mass x [0.962941, 1.12214, 1.1465, 1.06223]
  CoM XYZ mm [0.831907, 1.17711, -2.0201]
  mu_s [0.728342, 0.597748, 0.704749, 0.546287]
  mu_d [0.728342, 0.597748, 0.541926, 0.546287]
  Kp [steer,rear] [9.31056, 0]; Kd [0.183019, 0]
  spawn roll -0.00838106 rad; rate -0.188766 rad/s
  gyro bias XYZ rad/s [0.000118329, -2.55333e-05, 0.000110444]
  gravity bias YZ [-0.00132805, 0.00111189]
Six significant digits shown; full precision: sampled_dr_values.json.
```

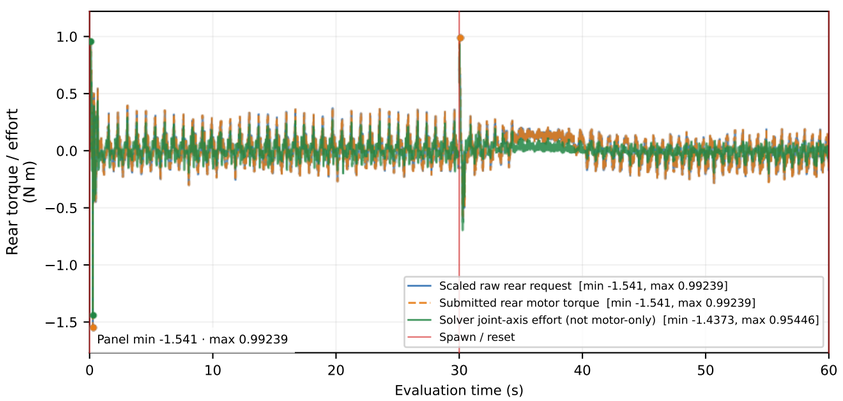
```
SAMPLED DR: env 49; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.913233, 1.1641, 1.12773, 0.91701]
  CoM XYZ mm [3.34854, 4.04615, -2.15763]
  mu_s [0.736077, 0.637675, 0.640016, 0.733515]
  mu_d [0.699455, 0.637675, 0.513141, 0.558392]
  Kp [steer,rear] [10.1476, 0]; Kd [0.213448, 0]
  spawn roll 0.0863189 rad; rate -0.0570891 rad/s
  gyro bias XYZ rad/s [0.000281771, -0.000166747, 0.000119675]
  gravity bias YZ [0.00174457, 0.00369669]
E2 [30.00, 30.20) s:
  mass x [1.05262, 1.19762, 0.845037, 0.981568]
  CoM XYZ mm [3.31378, -4.73576, -2.30997]
  mu_s [0.568073, 0.69316, 0.727426, 0.631146]
  mu_d [0.568073, 0.69316, 0.606358, 0.631146]
  Kp [steer,rear] [10.3401, 0]; Kd [0.216598, 0]
  spawn roll 0.0931377 rad; rate 0.193093 rad/s
  gyro bias XYZ rad/s [-0.000174902, -1.52907e-05, 5.03956e-05]
  gravity bias YZ [6.04822e-05, 0.00106008]
E3 [30.20, 60.00) s:
  mass x [0.903195, 0.901779, 1.18325, 0.942435]
  CoM XYZ mm [3.29046, 1.49534, -2.45754]
  mu_s [0.631146, 0.546287, 0.538434, 0.791209]
  mu_d [0.631146, 0.546287, 0.538434, 0.767936]
  Kp [steer,rear] [10.9304, 0]; Kd [0.211799, 0]
  spawn roll 0.0661296 rad; rate -0.0723594 rad/s
  gyro bias XYZ rad/s [1.84797e-06, 1.18338e-05, -0.00017462]
  gravity bias YZ [0.00631493, 0.00121862]
Six significant digits shown; full precision: sampled_dr_values.json.
```

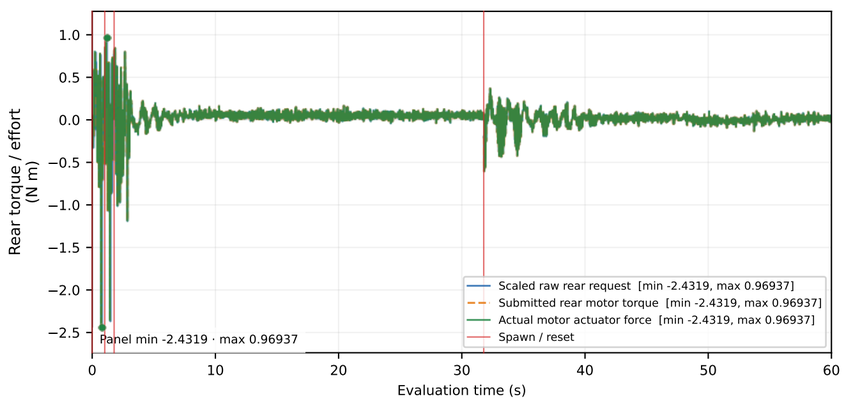
model_2500 · Demand versus actuation · env 50 · full 60 s

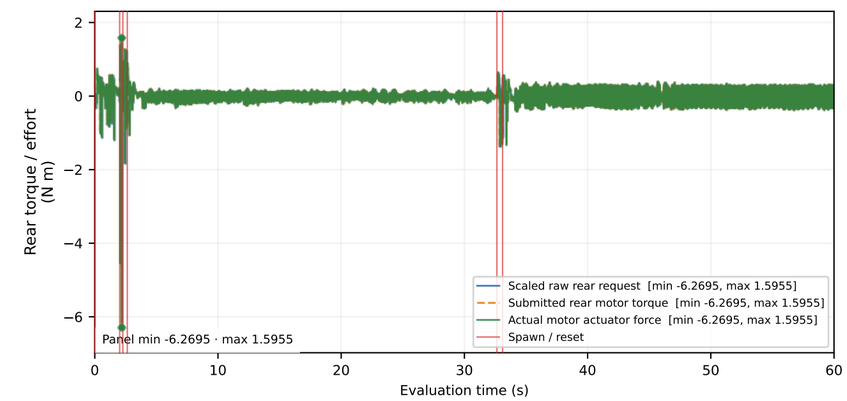












50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

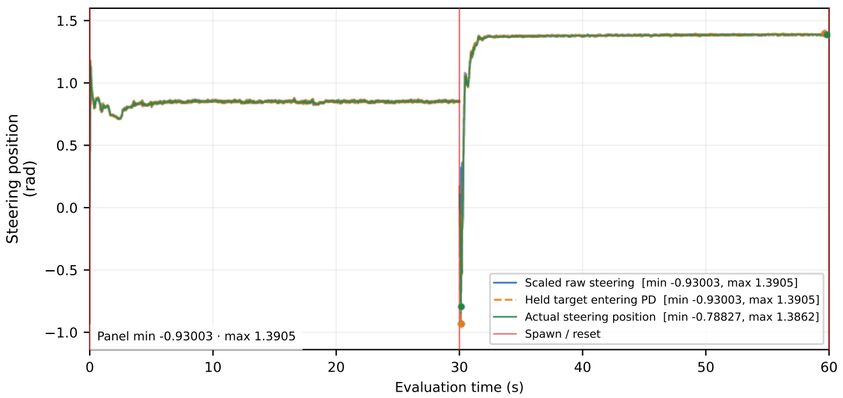
```
SAMPLED DR: env 50; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.15577, 0.969311, 0.993129, 0.980615]
  CoM XYZ mm [-2.54807, 1.30373, -2.17538]
  mu_s [0.751971, 0.764681, 0.764681, 0.693205]
  mu_d [0.541224, 0.764681, 0.764681, 0.693205]
  Kp [steer,rear] [10.8224, 0]; Kd [0.214074, 0]
  spawn roll -0.0884919 rad; rate -0.232642 rad/s
  gyro bias XYZ rad/s [0.000140697, 5.15835e-05, -0.000181121]
  gravity bias YZ [0.000233398, -0.000809242]
E2 [30.00, 60.00) s:
  mass x [1.18694, 0.806568, 0.949505, 0.879836]
  CoM XYZ mm [-3.80045, -2.67955, -3.76047]
  mu_s [0.785466, 0.532645, 0.760821, 0.760821]
  mu_d [0.52258, 0.532645, 0.670315, 0.670315]
  Kp [steer,rear] [10.3029, 0]; Kd [0.195031, 0]
  spawn roll 0.0895005 rad; rate 0.238192 rad/s
  gyro bias XYZ rad/s [-0.000127682, -0.000331672, 0.000413771]
  gravity bias YZ [-0.000251227, -0.00299106]
Six significant digits shown; full precision: sampled_dr_values.json.
```

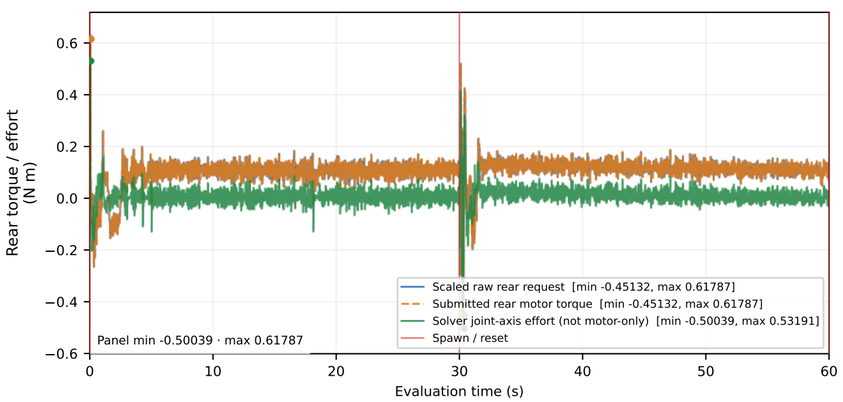
```
SAMPLED DR: env 50; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.02) s:
  mass x [1.04035, 0.830341, 0.940128, 0.933446]
  CoM XYZ mm [-4.62023, 3.57471, -4.64994]
  mu_s [0.631146, 0.732187, 0.597748, 0.631146]
  mu_d [0.631146, 0.631664, 0.597748, 0.631146]
  Kp [steer,rear] [9.08442, 0]; Kd [0.188041, 0]
  spawn roll 0.000687381 rad; rate -0.0492109 rad/s
  gyro bias XYZ rad/s [-4.09457e-05, -0.000282267, -0.00016859]
  gravity bias YZ [-0.000846323, -0.000773653]
E2 [1.02, 1.78) s:
  mass x [0.929787, 0.922862, 0.918625, 0.872758]
  CoM XYZ mm [2.83534, 1.67363, 0.0203466]
  mu_s [0.700521, 0.71155, 0.616244, 0.757579]
  mu_d [0.641329, 0.71155, 0.586498, 0.70921]
  Kp [steer,rear] [9.98066, 0]; Kd [0.195047, 0]
  spawn roll -0.0234883 rad; rate -0.223173 rad/s
  gyro bias XYZ rad/s [5.86512e-05, -0.000168819, 0.000113509]
  gravity bias YZ [0.000817212, -0.000122932]
E3 [1.78, 31.78) s:
  mass x [1.17386, 0.937473, 0.960115, 1.03836]
  CoM XYZ mm [-3.0154, 1.57505, 2.66425]
  mu_s [0.538976, 0.69316, 0.71008, 0.69316]
  mu_d [0.538976, 0.69316, 0.59371, 0.69316]
  Kp [steer,rear] [10.2724, 0]; Kd [0.180218, 0]
  spawn roll -0.0680822 rad; rate -0.181021 rad/s
  gyro bias XYZ rad/s [0.000169836, 6.95642e-05, 0.000131721]
  gravity bias YZ [0.000220411, 0.00153188]
E4 [31.78, 60.00) s:
  mass x [1.14809, 1.01048, 1.0009, 1.1621]
  CoM XYZ mm [4.30793, 2.27509, 0.90829]
  mu_s [0.69316, 0.640867, 0.791209, 0.749678]
  mu_d [0.69316, 0.556841, 0.767936, 0.741429]
  Kp [steer,rear] [9.29888, 0]; Kd [0.185282, 0]
  spawn roll -0.0135281 rad; rate 0.151796 rad/s
  gyro bias XYZ rad/s [0.000215499, 2.50955e-05, 0.000249005]
  gravity bias YZ [-0.0027542, -0.000682723]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 50; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 2.02) s:
  mass x [1.04035, 0.830341, 0.940128, 0.933446]
  CoM XYZ mm [-4.62023, 3.57471, -4.64994]
  mu_s [0.631146, 0.732187, 0.597748, 0.631146]
  mu_d [0.631146, 0.631664, 0.597748, 0.631146]
  Kp [steer,rear] [9.08442, 0]; Kd [0.188041, 0]
  spawn roll 0.000687381 rad; rate -0.0492109 rad/s
  gyro bias XYZ rad/s [-4.09457e-05, -0.000282267, -0.00016859]
  gravity bias YZ [-0.000846323, -0.000773653]
E2 [2.02, 2.28) s:
  mass x [1.15518, 0.810271, 0.828778, 1.07257]
  CoM XYZ mm [3.9666, 0.143293, -1.54585]
  mu_s [0.733515, 0.732187, 0.71155, 0.633024]
  mu_d [0.558392, 0.631664, 0.71155, 0.568172]
  Kp [steer,rear] [9.58315, 0]; Kd [0.202488, 0]
  spawn roll 0.0993319 rad; rate 0.00299045 rad/s
  gyro bias XYZ rad/s [-0.000284759, 4.29024e-05, -0.000236369]
  gravity bias YZ [-0.00225511, -0.0017409]
E3 [2.28, 2.64) s:
  mass x [0.809794, 0.840942, 0.932825, 1.19233]
  CoM XYZ mm [3.76137, 1.90411, 1.87976]
  mu_s [0.640016, 0.736077, 0.669571, 0.568073]
  mu_d [0.513141, 0.699455, 0.669571, 0.568073]
  Kp [steer,rear] [10.5795, 0]; Kd [0.185812, 0]
  spawn roll -0.0929147 rad; rate 0.0869282 rad/s
  gyro bias XYZ rad/s [0.000158452, -1.41046e-06, -5.29651e-05]
  gravity bias YZ [-0.00099225, -0.00054257]
E4 [2.64, 32.64) s:
  mass x [1.01306, 0.949765, 0.858973, 0.91838]
  CoM XYZ mm [-4.29277, -3.43135, 4.84826]
  mu_s [0.728342, 0.633024, 0.69316, 0.757579]
  mu_d [0.728342, 0.568172, 0.69316, 0.70921]
  Kp [steer,rear] [10.2756, 0]; Kd [0.206824, 0]
  spawn roll -0.0734395 rad; rate -0.222717 rad/s
  gyro bias XYZ rad/s [0.000155822, 0.000201897, 5.28504e-05]
  gravity bias YZ [-0.00252984, 0.000504211]
E5 [32.64, 33.10) s:
  mass x [0.950504, 0.884692, 0.910554, 1.06995]
  CoM XYZ mm [0.345044, 0.532505, 2.15373]
  mu_s [0.71008, 0.559972, 0.733515, 0.728342]
  mu_d [0.59371, 0.502209, 0.558392, 0.728342]
  Kp [steer,rear] [9.56995, 0]; Kd [0.201848, 0]
  spawn roll -0.0548667 rad; rate -0.0165733 rad/s
  gyro bias XYZ rad/s [-2.62859e-06, 9.29744e-05, -0.000120543]
  gravity bias YZ [0.000714771, -0.00144499]
E6 [33.10, 60.00) s:
  mass x [1.09967, 1.09061, 1.05979, 1.16773]
  CoM XYZ mm [3.92281, -0.491529, 0.452252]
  mu_s [0.791209, 0.541939, 0.732187, 0.749678]
  mu_d [0.767936, 0.534359, 0.631664, 0.741429]
  Kp [steer,rear] [10.32, 0]; Kd [0.195267, 0]
  spawn roll -0.0463786 rad; rate 0.00287988 rad/s
  gyro bias XYZ rad/s [-0.000251113, 0.000130964, 0.000205204]
  gravity bias YZ [-0.000343513, 0.0016595]
Six significant digits shown; full precision: sampled_dr_values.json.
```

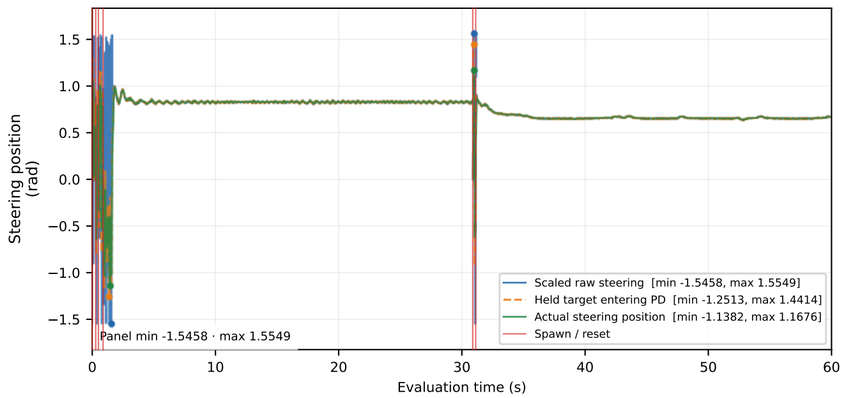
model_2500 · Demand versus actuation · env 51 · full 60 s

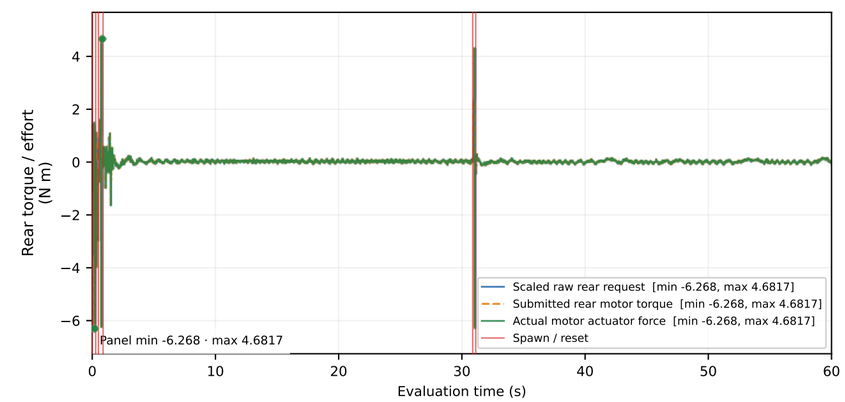
Isaac Lab · physics/PD 400 Hz · env 51
DR · 0.00–60.00 s



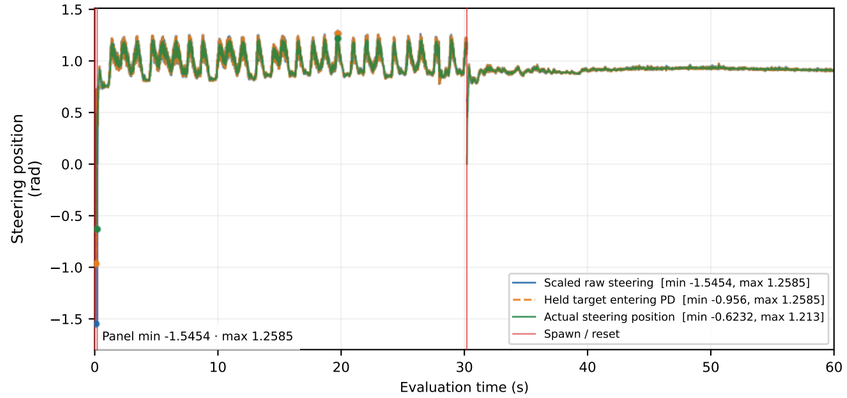


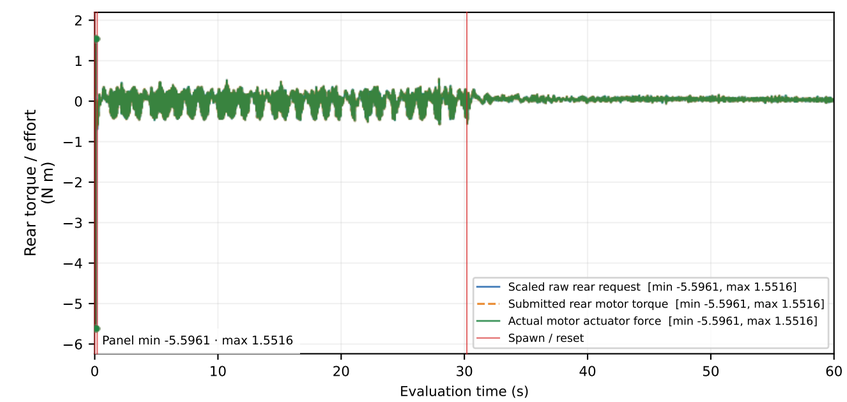
MuJoCo · physics/PD 400 Hz · env 51
DR · 0.00–60.00 s





MuJoCo · physics/PD 800 Hz · env 51
DR · 0.00–60.00 s





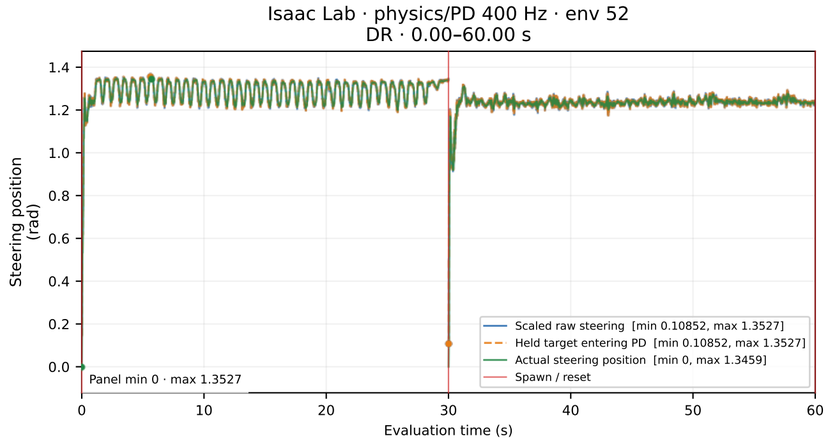
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

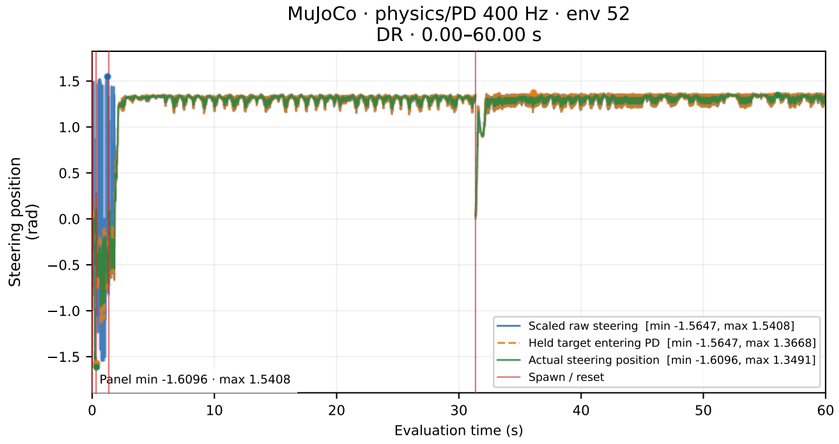
```
SAMPLED DR: env 51; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.87433, 1.19755, 0.963269, 0.971932]
  CoM XYZ mm [-1.10874, 4.205, -3.25464]
  mu_s [0.533455, 0.787792, 0.659445, 0.674963]
  mu_d [0.533455, 0.617134, 0.547619, 0.601294]
  Kp [steer,rear] [10.5429, 0]; Kd [0.189101, 0]
  spawn roll 0.0687147 rad; rate -0.105067 rad/s
  gyro bias XYZ rad/s [0.000170898, -5.42507e-05, 0.000132994]
  gravity bias YZ [0.00215066, -0.00110346]
E2 [30.00, 60.00) s:
  mass x [1.19788, 1.07216, 1.01968, 0.875061]
  CoM XYZ mm [2.23897, -4.09974, -4.60489]
  mu_s [0.533455, 0.598343, 0.751971, 0.632409]
  mu_d [0.533455, 0.598343, 0.541224, 0.589076]
  Kp [steer,rear] [10.859, 0]; Kd [0.219522, 0]
  spawn roll -0.0552671 rad; rate 0.156633 rad/s
  gyro bias XYZ rad/s [-8.94421e-05, 2.81277e-05, -0.000141213]
  gravity bias YZ [0.000624014, 0.00274033]
Six significant digits shown; full precision: sampled_dr_values.json.
```

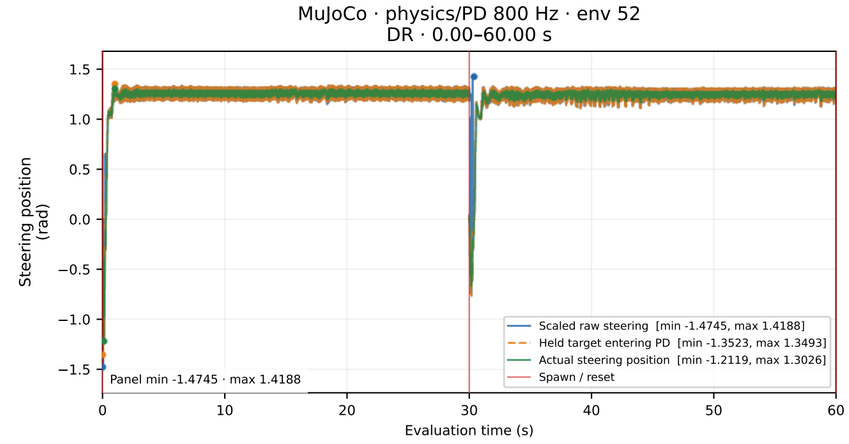
```
SAMPLED DR: env 51; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [1.1942, 1.01727, 0.983743, 0.973919]
  CoM XYZ mm [1.6316, -4.43244, 4.72033]
  mu_s [0.69316, 0.546287, 0.546287, 0.704749]
  mu_d [0.69316, 0.546287, 0.546287, 0.541926]
  Kp [steer,rear] [9.06551, 0]; Kd [0.21998, 0]
  spawn roll 0.0866621 rad; rate -0.22737 rad/s
  gyro bias XYZ rad/s [0.000142895, -0.000194485, -0.000107844]
  gravity bias YZ [0.00221892, 0.00273361]
E2 [0.28, 0.50) s:
  mass x [0.996803, 1.00338, 0.949832, 0.95773]
  CoM XYZ mm [-4.12259, -2.14861, -0.173559]
  mu_s [0.736077, 0.631146, 0.728342, 0.616244]
  mu_d [0.699455, 0.631146, 0.728342, 0.586498]
  Kp [steer,rear] [10.9851, 0]; Kd [0.182833, 0]
  spawn roll 0.0976148 rad; rate 0.101993 rad/s
  gyro bias XYZ rad/s [0.000105736, -0.000221656, -0.000179136]
  gravity bias YZ [-0.000576773, 0.000886982]
E3 [0.50, 0.88) s:
  mass x [1.05313, 1.11235, 0.994234, 0.868535]
  CoM XYZ mm [2.37178, -1.56825, -3.62737]
  mu_s [0.749678, 0.704749, 0.546287, 0.538434]
  mu_d [0.741429, 0.541926, 0.546287, 0.538434]
  Kp [steer,rear] [10.6829, 0]; Kd [0.218654, 0]
  spawn roll 0.0716195 rad; rate -0.117495 rad/s
  gyro bias XYZ rad/s [0.000107159, -9.19193e-05, 5.4211e-05]
  gravity bias YZ [0.00365437, 0.00200592]
E4 [0.88, 30.88) s:
  mass x [1.15023, 1.09454, 1.18487, 0.981418]
  CoM XYZ mm [1.46313, 4.18747, -2.39991]
  mu_s [0.640867, 0.633024, 0.640867, 0.611239]
  mu_d [0.556841, 0.568172, 0.556841, 0.611239]
  Kp [steer,rear] [9.64987, 0]; Kd [0.181477, 0]
  spawn roll -0.0252405 rad; rate 0.0123658 rad/s
  gyro bias XYZ rad/s [-1.4485e-06, 0.000125752, -4.57125e-05]
  gravity bias YZ [0.000808513, 0.00196315]
E5 [30.88, 31.12) s:
  mass x [1.05561, 0.952827, 0.80234, 0.958497]
  CoM XYZ mm [-2.01471, -3.74206, -4.28537]
  mu_s [0.633024, 0.71008, 0.669571, 0.597748]
  mu_d [0.568172, 0.59371, 0.669571, 0.597748]
  Kp [steer,rear] [9.2209, 0]; Kd [0.214402, 0]
  spawn roll 0.0690499 rad; rate -0.0117621 rad/s
  gyro bias XYZ rad/s [-4.94755e-05, -0.000266939, 0.000412373]
  gravity bias YZ [-0.00155495, -0.00130275]
E6 [31.12, 60.00) s:
  mass x [1.01532, 0.931921, 1.07484, 1.07977]
  CoM XYZ mm [-4.74891, 4.88739, -1.84585]
  mu_s [0.538976, 0.704749, 0.637675, 0.757579]
  mu_d [0.538976, 0.541926, 0.637675, 0.70921]
  Kp [steer,rear] [9.01141, 0]; Kd [0.190206, 0]
  spawn roll 0.0474584 rad; rate 0.0501261 rad/s
  gyro bias XYZ rad/s [-0.000107537, 0.000138557, -3.35874e-05]
  gravity bias YZ [-0.00247383, -0.00157511]
Six significant digits shown; full precision: sampled_dr_values.json.
```

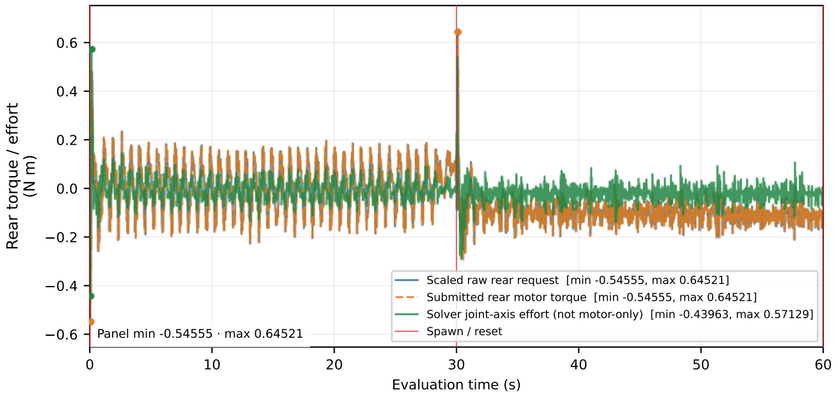
```
SAMPLED DR: env 51; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.20) s:
  mass x [1.1942, 1.01727, 0.983743, 0.973919]
  CoM XYZ mm [1.6316, -4.43244, 4.72033]
  mu_s [0.69316, 0.546287, 0.546287, 0.704749]
  mu_d [0.69316, 0.546287, 0.546287, 0.541926]
  Kp [steer,rear] [9.06551, 0]; Kd [0.21998, 0]
  spawn roll 0.0866621 rad; rate -0.22737 rad/s
  gyro bias XYZ rad/s [0.000142895, -0.000194485, -0.000107844]
  gravity bias YZ [0.00221892, 0.00273361]
E2 [0.20, 30.20) s:
  mass x [0.892959, 1.08917, 0.981928, 1.01878]
  CoM XYZ mm [3.27123, 2.47138, 2.76163]
  mu_s [0.611239, 0.732187, 0.700521, 0.728342]
  mu_d [0.611239, 0.631664, 0.641329, 0.728342]
  Kp [steer,rear] [10.3149, 0]; Kd [0.193014, 0]
  spawn roll -0.0296293 rad; rate 0.215667 rad/s
  gyro bias XYZ rad/s [0.000240257, 0.00035221, 0.000259583]
  gravity bias YZ [0.000904406, 0.0010964]
E3 [30.20, 60.00) s:
  mass x [0.971615, 0.966917, 0.991726, 0.828326]
  CoM XYZ mm [-3.34545, 2.40684, -1.81808]
  mu_s [0.748289, 0.732187, 0.704749, 0.528253]
  mu_d [0.689499, 0.631664, 0.541926, 0.528253]
  Kp [steer,rear] [9.8693, 0]; Kd [0.186607, 0]
  spawn roll -0.00592788 rad; rate 0.207095 rad/s
  gyro bias XYZ rad/s [-5.1523e-05, 0.0002882, 0.00029299]
  gravity bias YZ [0.000287504, -0.00250863]
Six significant digits shown; full precision: sampled_dr_values.json.
```

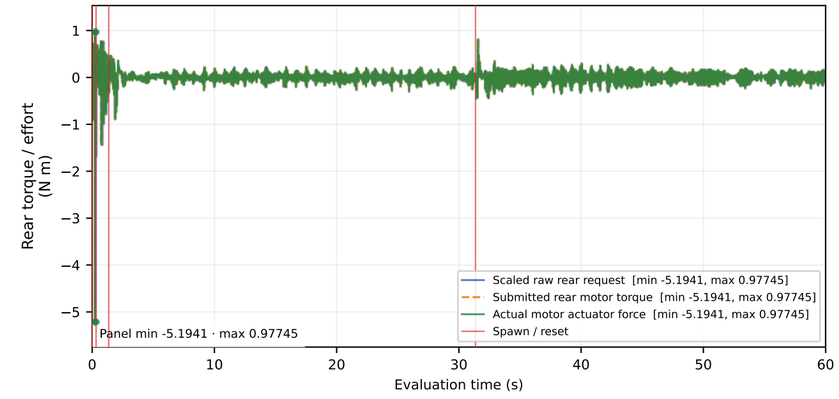
model_2500 · Demand versus actuation · env 52 · full 60 s

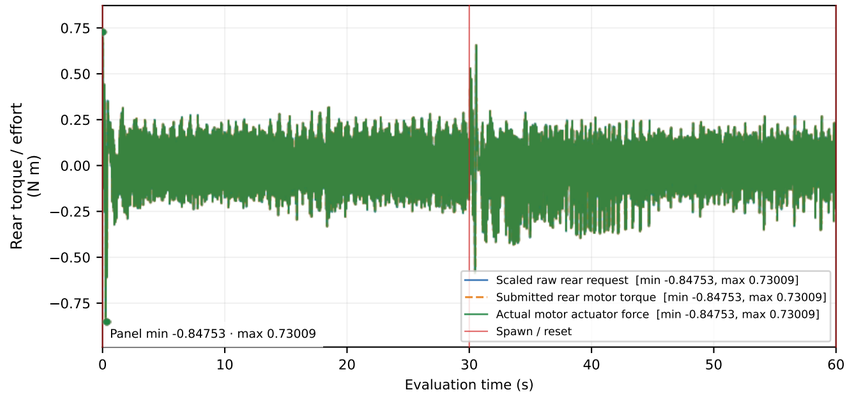












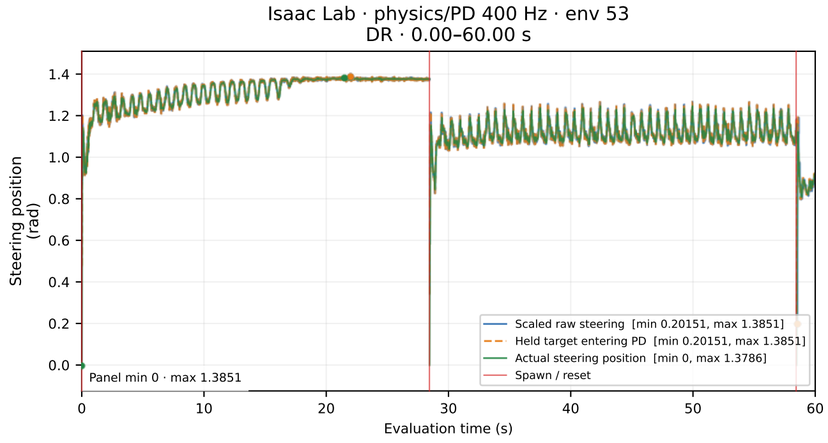
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

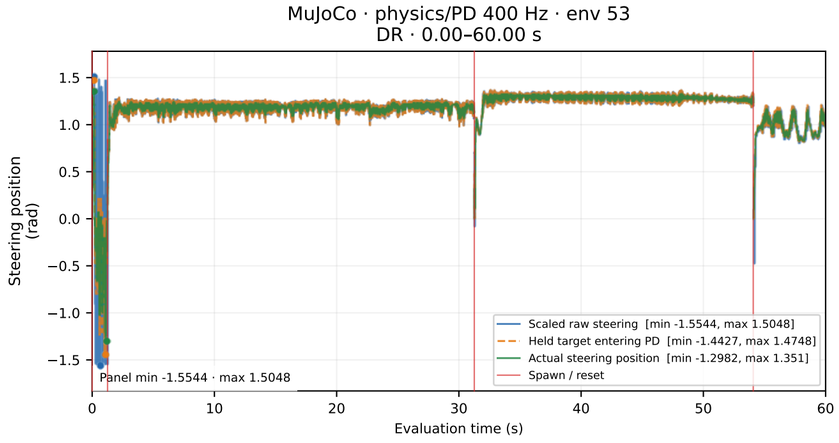
```
SAMPLED DR: env 52; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.01143, 0.921674, 0.904912, 0.988041]
  CoM XYZ mm [-1.64346, -2.42152, -3.33379]
  mu_s [0.532645, 0.751971, 0.628821, 0.751971]
  mu_d [0.532645, 0.541224, 0.628821, 0.541224]
  Kp [steer,rear] [9.6431, 0]; Kd [0.197862, 0]
  spawn roll -0.0318425 rad; rate 0.0760781 rad/s
  gyro bias XYZ rad/s [-8.25792e-05, -5.09338e-05, -0.000109599]
  gravity bias YZ [-0.0010959, -0.00106676]
E2 [30.00, 60.00) s:
  mass x [0.978785, 1.12238, 0.900565, 0.925982]
  CoM XYZ mm [4.98074, -3.23945, 1.40122]
  mu_s [0.751971, 0.559809, 0.628821, 0.713129]
  mu_d [0.541224, 0.559809, 0.628821, 0.713129]
  Kp [steer,rear] [10.7297, 0]; Kd [0.201636, 0]
  spawn roll 0.0360942 rad; rate -0.145563 rad/s
  gyro bias XYZ rad/s [-0.000104037, -4.83557e-05, 5.78648e-05]
  gravity bias YZ [-0.00165757, 5.49024e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

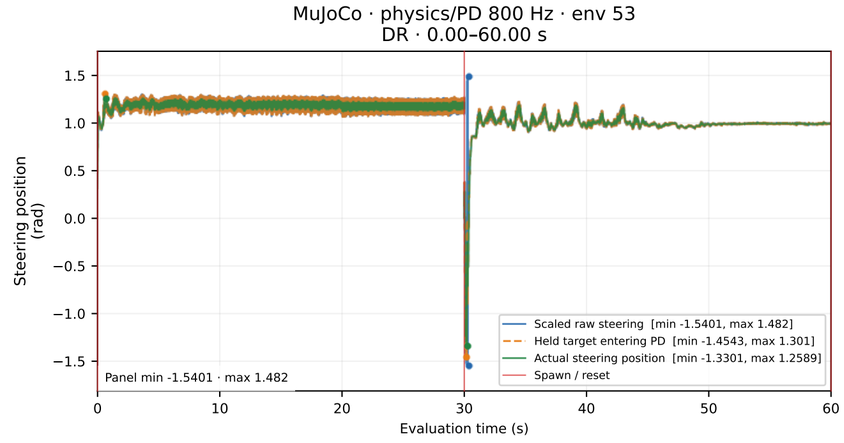
```
SAMPLED DR: env 52; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.32) s:
  mass x [0.895868, 1.11548, 1.08882, 0.818311]
  CoM XYZ mm [-1.70543, -2.37887, 0.642735]
  mu_s [0.69316, 0.637675, 0.640016, 0.748289]
  mu_d [0.69316, 0.637675, 0.513141, 0.689499]
  Kp [steer,rear] [10.9442, 0]; Kd [0.212329, 0]
  spawn roll -0.0656004 rad; rate -0.0835782 rad/s
  gyro bias XYZ rad/s [-4.7977e-05, 6.89662e-05, -9.91746e-05]
  gravity bias YZ [-0.000248333, -0.000571064]
E2 [0.32, 1.36) s:
  mass x [0.902628, 1.1733, 0.854954, 0.965253]
  CoM XYZ mm [0.700614, 4.80222, 4.26717]
  mu_s [0.640016, 0.732187, 0.700521, 0.666375]
  mu_d [0.513141, 0.631664, 0.641329, 0.519145]
  Kp [steer,rear] [10.0627, 0]; Kd [0.197855, 0]
  spawn roll 0.0103001 rad; rate -0.208955 rad/s
  gyro bias XYZ rad/s [9.74961e-05, 0.000102484, 0.000323916]
  gravity bias YZ [0.0003707, 0.0020176]
E3 [1.36, 31.36) s:
  mass x [1.19162, 1.19224, 1.00279, 1.1625]
  CoM XYZ mm [-1.02146, -2.66206, -1.10231]
  mu_s [0.669571, 0.559972, 0.637675, 0.71155]
  mu_d [0.669571, 0.502209, 0.637675, 0.71155]
  Kp [steer,rear] [9.80641, 0]; Kd [0.200827, 0]
  spawn roll -0.0404535 rad; rate 0.114395 rad/s
  gyro bias XYZ rad/s [-4.05207e-05, -2.45261e-06, -3.14968e-05]
  gravity bias YZ [0.00122725, -0.000385151]
E4 [31.36, 60.00) s:
  mass x [0.996757, 0.962115, 1.19327, 0.997593]
  CoM XYZ mm [3.8777, -2.88413, -0.403296]
  mu_s [0.723429, 0.757579, 0.700521, 0.528253]
  mu_d [0.723429, 0.70921, 0.641329, 0.528253]
  Kp [steer,rear] [9.35724, 0]; Kd [0.208195, 0]
  spawn roll -0.0178964 rad; rate 0.006131 rad/s
  gyro bias XYZ rad/s [0.000110999, -0.000215221, -9.27952e-05]
  gravity bias YZ [0.00187409, -6.57911e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

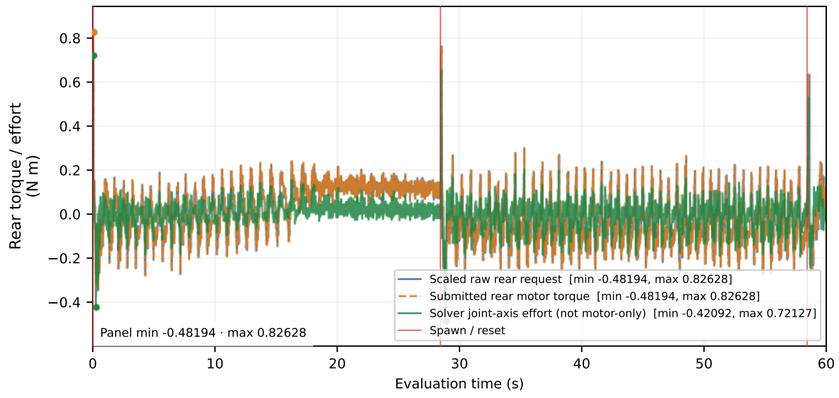
```
SAMPLED DR: env 52; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.895868, 1.11548, 1.08882, 0.818311]
  CoM XYZ mm [-1.70543, -2.37887, 0.642735]
  mu_s [0.69316, 0.637675, 0.640016, 0.748289]
  mu_d [0.69316, 0.637675, 0.513141, 0.689499]
  Kp [steer,rear] [10.9442, 0]; Kd [0.212329, 0]
  spawn roll -0.0656004 rad; rate -0.0835782 rad/s
  gyro bias XYZ rad/s [-4.7977e-05, 6.89662e-05, -9.91746e-05]
  gravity bias YZ [-0.000248333, -0.000571064]
E2 [30.00, 60.00) s:
  mass x [1.13978, 0.995018, 0.97675, 0.876827]
  CoM XYZ mm [1.76979, -1.93484, 3.17605]
  mu_s [0.597748, 0.637675, 0.633024, 0.611239]
  mu_d [0.597748, 0.637675, 0.568172, 0.611239]
  Kp [steer,rear] [10.201, 0]; Kd [0.201236, 0]
  spawn roll -0.0432077 rad; rate 0.0697385 rad/s
  gyro bias XYZ rad/s [-0.000407284, 0.000293345, 0.000263711]
  gravity bias YZ [-0.000910457, 0.000317174]
Six significant digits shown; full precision: sampled_dr_values.json.
```

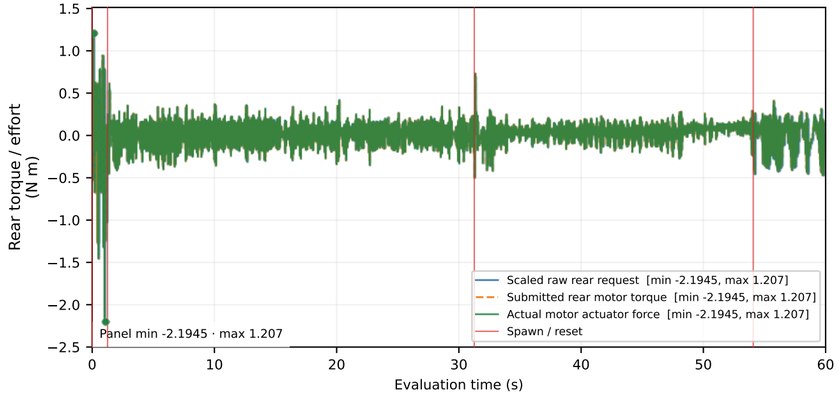
model_2500 · Demand versus actuation · env 53 · full 60 s

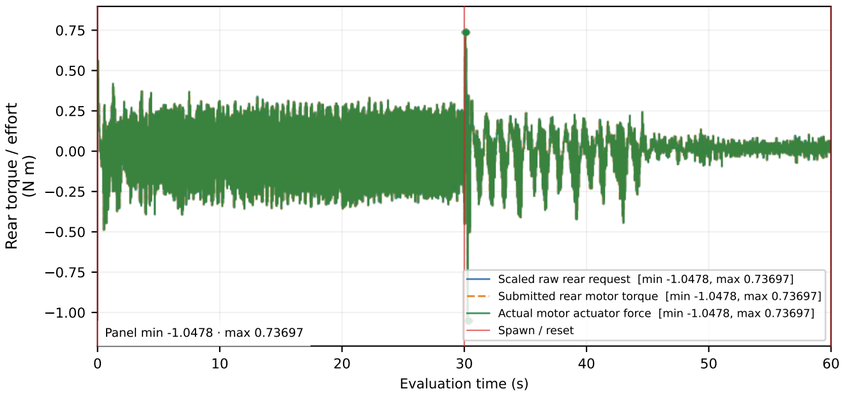












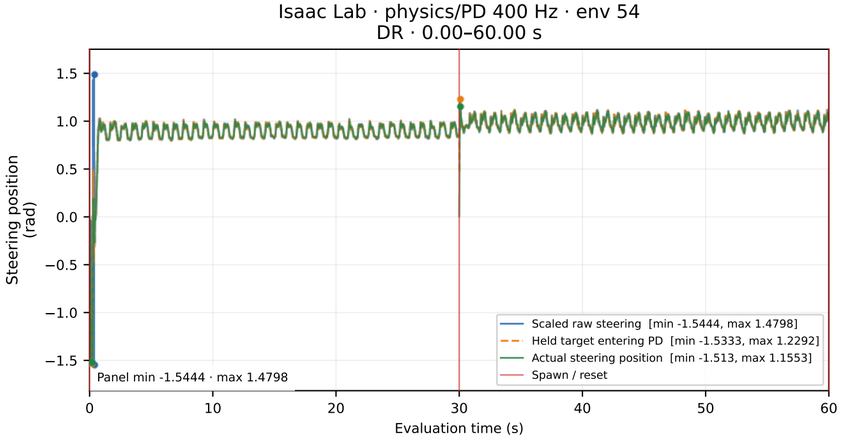
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

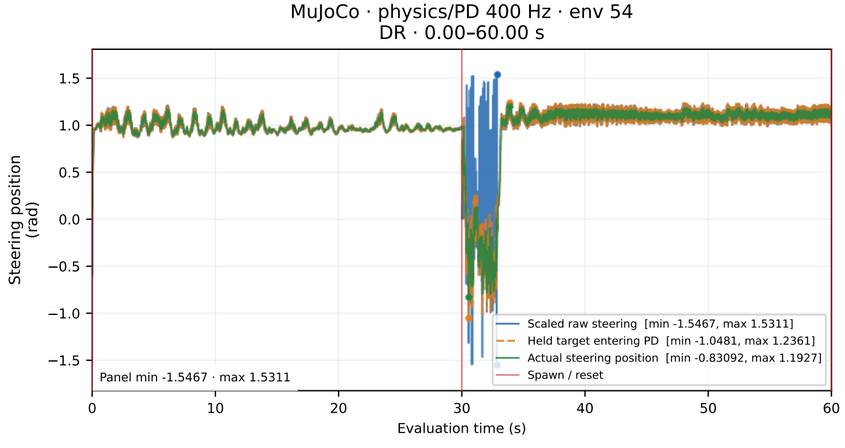
```
SAMPLED DR: env 53; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 28.44) s:
  mass x [0.937191, 1.19344, 0.855895, 1.14525]
  CoM XYZ mm [-1.96986, -2.81425, 2.27009]
  mu_s [0.774409, 0.539956, 0.774409, 0.785466]
  mu_d [0.561095, 0.539956, 0.561095, 0.52258]
  Kp [steer,rear] [9.10881, 0]; Kd [0.200427, 0]
  spawn roll 0.0774201 rad; rate 0.199795 rad/s
  gyro bias XYZ rad/s [-2.1234e-05, 7.10412e-05, 0.000266473]
  gravity bias YZ [0.00113377, 0.00341815]
E2 [28.44, 58.44) s:
  mass x [1.08785, 0.933505, 0.894214, 0.904942]
  CoM XYZ mm [2.82797, -0.615619, 0.79608]
  mu_s [0.531594, 0.774409, 0.693205, 0.531594]
  mu_d [0.531594, 0.561095, 0.693205, 0.531594]
  Kp [steer,rear] [9.15879, 0]; Kd [0.188207, 0]
  spawn roll 0.065437 rad; rate -0.16356 rad/s
  gyro bias XYZ rad/s [4.0565e-05, -0.000219733, 6.90915e-06]
  gravity bias YZ [-0.00324229, -0.000688782]
E3 [58.44, 60.00) s:
  mass x [1.02962, 1.16839, 1.18665, 0.912459]
  CoM XYZ mm [-4.35281, 3.14168, -1.23222]
  mu_s [0.741039, 0.659445, 0.713129, 0.647391]
  mu_d [0.614405, 0.547619, 0.713129, 0.647391]
  Kp [steer,rear] [10.5804, 0]; Kd [0.211518, 0]
  spawn roll 0.0621697 rad; rate -0.189606 rad/s
  gyro bias XYZ rad/s [0.000247501, 3.50317e-05, 0.000232372]
  gravity bias YZ [0.00033728, 0.00111157]
Six significant digits shown; full precision: sampled_dr_values.json.
```

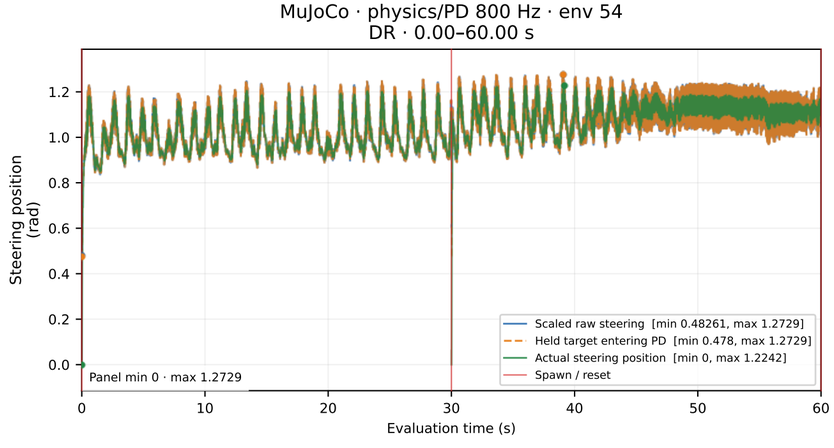
```
SAMPLED DR: env 53; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.26) s:
  mass x [1.01658, 1.16532, 0.890464, 0.952267]
  CoM XYZ mm [-0.896499, -0.23245, -3.38234]
  mu_s [0.616244, 0.704749, 0.791209, 0.559972]
  mu_d [0.586498, 0.541926, 0.767936, 0.502209]
  Kp [steer,rear] [9.83361, 0]; Kd [0.199999, 0]
  spawn roll 0.0480047 rad; rate 0.0813598 rad/s
  gyro bias XYZ rad/s [-1.68078e-05, -0.000542628, 0.000220491]
  gravity bias YZ [0.00110247, 0.000413204]
E2 [1.26, 31.26) s:
  mass x [0.854132, 1.05172, 0.994098, 1.09104]
  CoM XYZ mm [-1.3453, -0.772613, 3.85896]
  mu_s [0.568073, 0.723429, 0.611239, 0.541939]
  mu_d [0.568073, 0.723429, 0.611239, 0.534359]
  Kp [steer,rear] [10.1132, 0]; Kd [0.21786, 0]
  spawn roll -9.27329e-05 rad; rate -0.0141128 rad/s
  gyro bias XYZ rad/s [6.8371e-05, -7.18617e-05, 0.000189908]
  gravity bias YZ [-0.00102904, -0.000470824]
E3 [31.26, 54.08) s:
  mass x [0.922679, 0.983297, 0.956886, 1.0459]
  CoM XYZ mm [4.99604, -4.88986, 3.77766]
  mu_s [0.559972, 0.704749, 0.568073, 0.538976]
  mu_d [0.502209, 0.541926, 0.568073, 0.538976]
  Kp [steer,rear] [10.7575, 0]; Kd [0.205237, 0]
  spawn roll 0.0373654 rad; rate 0.115207 rad/s
  gyro bias XYZ rad/s [0.000234425, 6.02528e-05, -0.000494244]
  gravity bias YZ [-0.00335598, 0.00117259]
E4 [54.08, 60.00) s:
  mass x [0.953553, 1.12825, 1.05418, 0.953149]
  CoM XYZ mm [3.20866, 2.32445, 2.39002]
  mu_s [0.538434, 0.736077, 0.631146, 0.568073]
  mu_d [0.538434, 0.699455, 0.631146, 0.568073]
  Kp [steer,rear] [10.0327, 0]; Kd [0.184673, 0]
  spawn roll 0.0341392 rad; rate 0.122202 rad/s
  gyro bias XYZ rad/s [-0.000102103, -0.000161652, -0.00031826]
  gravity bias YZ [0.00161921, 4.34852e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

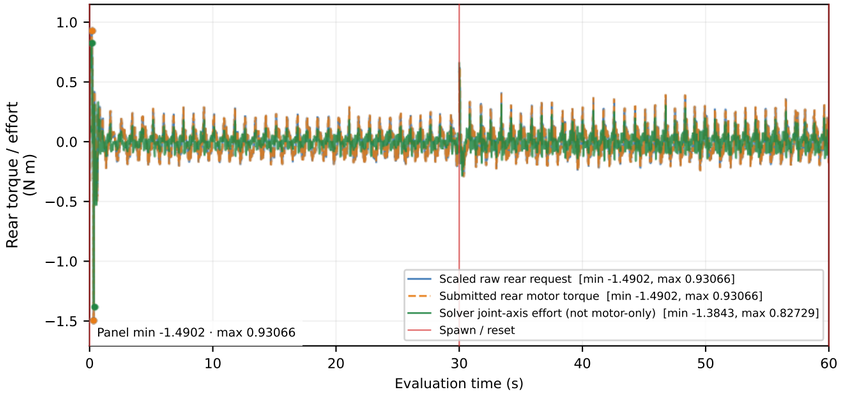
```
SAMPLED DR: env 53; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.01658, 1.16532, 0.890464, 0.952267]
  CoM XYZ mm [-0.896499, -0.23245, -3.38234]
  mu_s [0.616244, 0.704749, 0.791209, 0.559972]
  mu_d [0.586498, 0.541926, 0.767936, 0.502209]
  Kp [steer,rear] [9.83361, 0]; Kd [0.199999, 0]
  spawn roll 0.0480047 rad; rate 0.0813598 rad/s
  gyro bias XYZ rad/s [-1.68078e-05, -0.000542628, 0.000220491]
  gravity bias YZ [0.00110247, 0.000413204]
E2 [30.00, 60.00) s:
  mass x [1.15391, 1.11127, 0.962362, 1.17442]
  CoM XYZ mm [-4.7896, 0.566434, 2.75297]
  mu_s [0.728342, 0.69316, 0.528253, 0.71155]
  mu_d [0.728342, 0.69316, 0.528253, 0.71155]
  Kp [steer,rear] [10.9894, 0]; Kd [0.204184, 0]
  spawn roll -0.0670953 rad; rate 0.171254 rad/s
  gyro bias XYZ rad/s [-0.000121232, -5.29606e-05, 0.000156749]
  gravity bias YZ [-0.000861907, -0.0036078]
Six significant digits shown; full precision: sampled_dr_values.json.
```

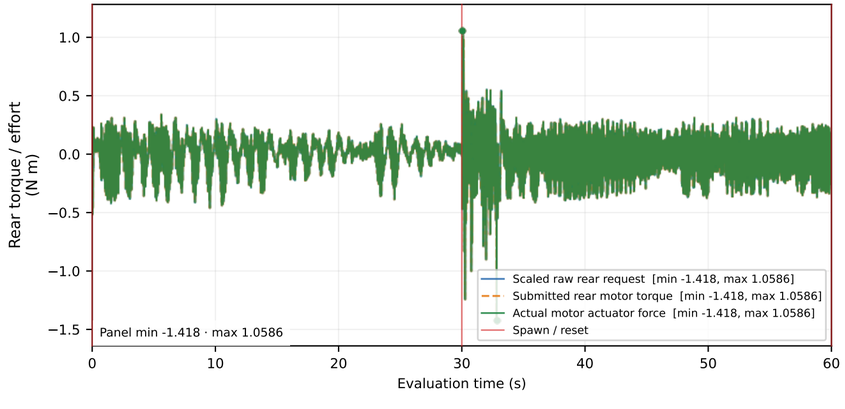
model_2500 · Demand versus actuation · env 54 · full 60 s

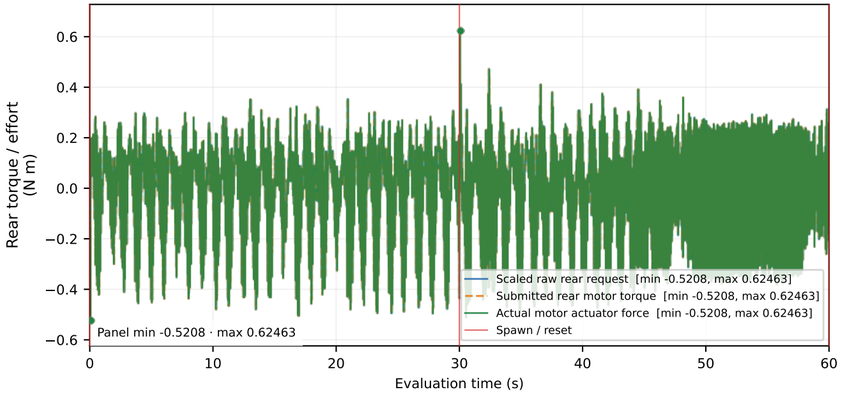












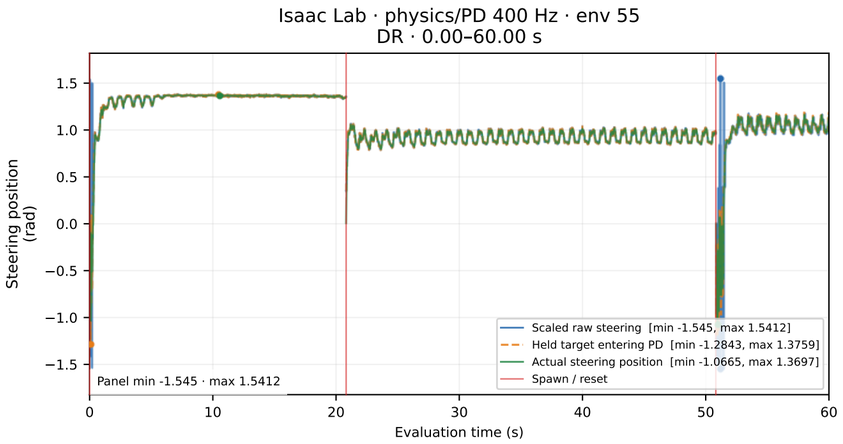
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

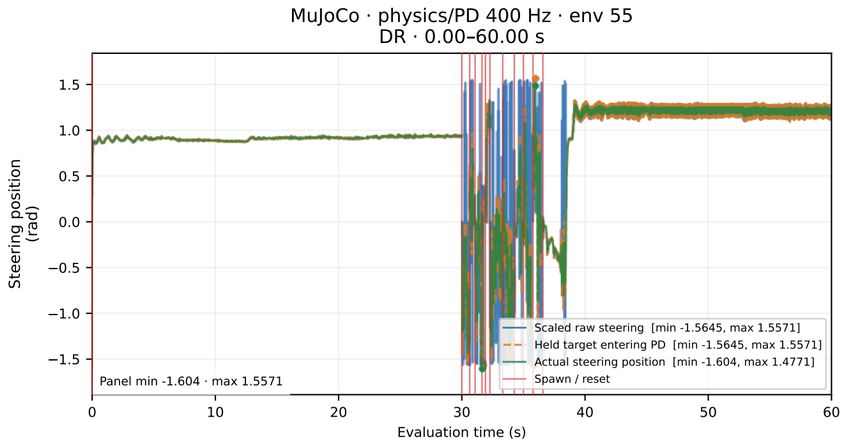
```
SAMPLED DR: env 54; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.10934, 1.19379, 0.825476, 1.17256]
  CoM XYZ mm [2.97143, 2.55803, 3.43199]
  mu_s [0.560535, 0.703711, 0.726175, 0.681711]
  mu_d [0.560535, 0.703711, 0.558574, 0.611756]
  Kp [steer,rear] [10.0082, 0]; Kd [0.186213, 0]
  spawn roll -0.0964281 rad; rate 0.181293 rad/s
  gyro bias XYZ rad/s [-0.000101903, -9.48728e-05, 6.15729e-05]
  gravity bias YZ [-0.000767394, 0.000607192]
E2 [30.00, 60.00) s:
  mass x [0.921562, 0.915659, 0.964448, 0.957485]
  CoM XYZ mm [-0.282958, 0.988118, -3.6774]
  mu_s [0.751971, 0.584424, 0.547743, 0.726175]
  mu_d [0.541224, 0.584424, 0.547743, 0.558574]
  Kp [steer,rear] [9.502, 0]; Kd [0.182184, 0]
  spawn roll 0.0621586 rad; rate 0.244119 rad/s
  gyro bias XYZ rad/s [-0.000509188, -3.44135e-05, 4.87007e-06]
  gravity bias YZ [-0.0040903, -0.000794371]
Six significant digits shown; full precision: sampled_dr_values.json.
```

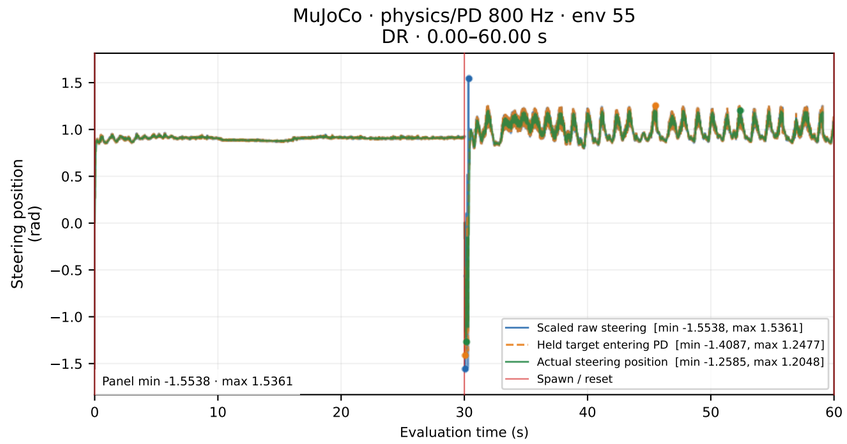
```
SAMPLED DR: env 54; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09167, 1.09356, 1.01932, 0.995919]
  CoM XYZ mm [0.318897, 1.84907, -4.07307]
  mu_s [0.666375, 0.700521, 0.723429, 0.541939]
  mu_d [0.519145, 0.641329, 0.723429, 0.534359]
  Kp [steer,rear] [9.46515, 0]; Kd [0.187392, 0]
  spawn roll 0.0116196 rad; rate 0.0807588 rad/s
  gyro bias XYZ rad/s [6.48641e-05, -8.75067e-05, 0.000224428]
  gravity bias YZ [0.0010855, -0.00040928]
E2 [30.00, 60.00) s:
  mass x [1.09167, 0.850294, 1.03901, 1.13524]
  CoM XYZ mm [2.66514, 1.43688, 1.52467]
  mu_s [0.732187, 0.669571, 0.559972, 0.597748]
  mu_d [0.631664, 0.669571, 0.502209, 0.597748]
  Kp [steer,rear] [9.70921, 0]; Kd [0.203106, 0]
  spawn roll 0.0644941 rad; rate -0.00531168 rad/s
  gyro bias XYZ rad/s [0.000128906, 0.000211222, -0.000220637]
  gravity bias YZ [0.00353561, 0.000974373]
Six significant digits shown; full precision: sampled_dr_values.json.
```

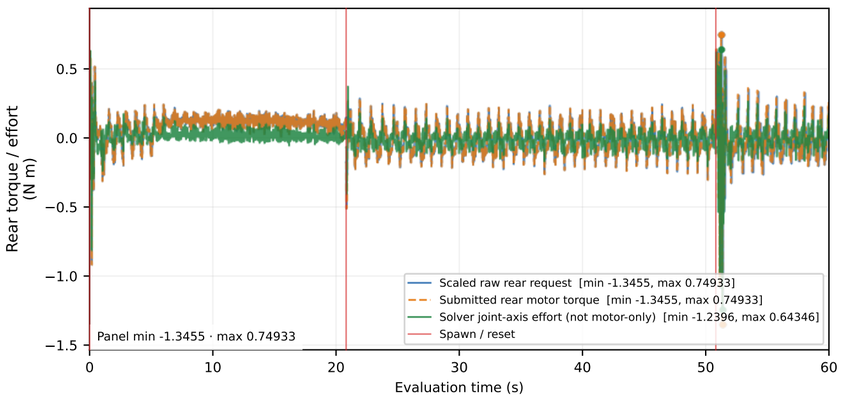
```
SAMPLED DR: env 54; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09167, 1.09356, 1.01932, 0.995919]
  CoM XYZ mm [0.318897, 1.84907, -4.07307]
  mu_s [0.666375, 0.700521, 0.723429, 0.541939]
  mu_d [0.519145, 0.641329, 0.723429, 0.534359]
  Kp [steer,rear] [9.46515, 0]; Kd [0.187392, 0]
  spawn roll 0.0116196 rad; rate 0.0807588 rad/s
  gyro bias XYZ rad/s [6.48641e-05, -8.75067e-05, 0.000224428]
  gravity bias YZ [0.0010855, -0.00040928]
E2 [30.00, 60.00) s:
  mass x [1.09167, 0.850294, 1.03901, 1.13524]
  CoM XYZ mm [2.66514, 1.43688, 1.52467]
  mu_s [0.732187, 0.669571, 0.559972, 0.597748]
  mu_d [0.631664, 0.669571, 0.502209, 0.597748]
  Kp [steer,rear] [9.70921, 0]; Kd [0.203106, 0]
  spawn roll 0.0644941 rad; rate -0.00531168 rad/s
  gyro bias XYZ rad/s [0.000128906, 0.000211222, -0.000220637]
  gravity bias YZ [0.00353561, 0.000974373]
Six significant digits shown; full precision: sampled_dr_values.json.
```

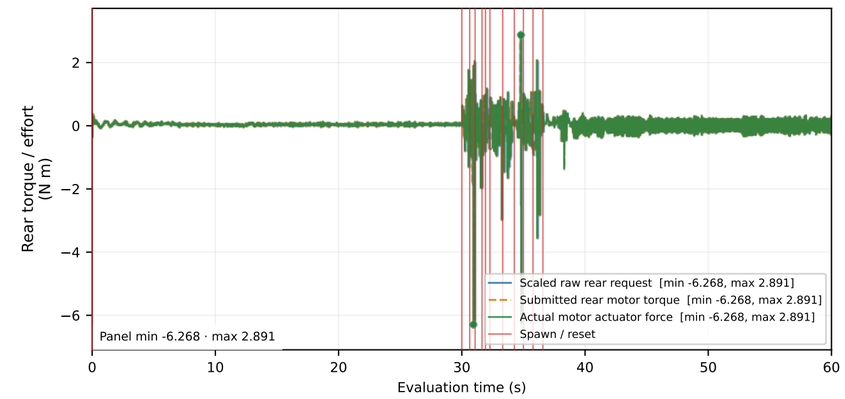
model_2500 · Demand versus actuation · env 55 · full 60 s

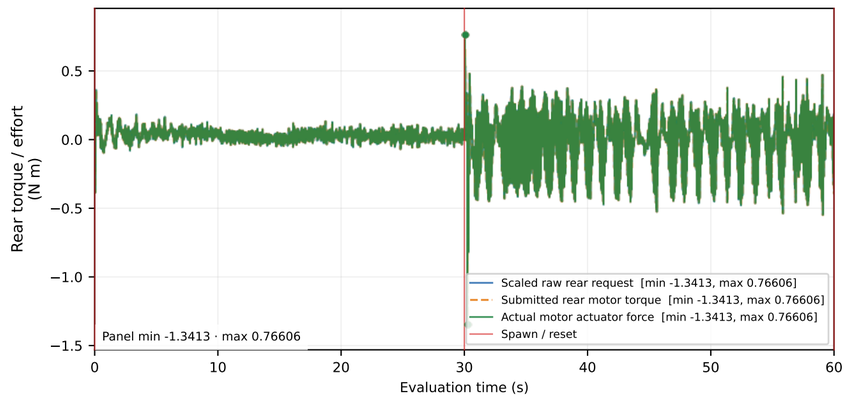












50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 55; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 20.82) s:
  mass x [0.963485, 1.07764, 1.18683, 0.820207]
  CoM XYZ mm [-3.90576, -2.59108, 0.0577271]
  mu_s [0.628821, 0.560535, 0.760821, 0.647391]
  mu_d [0.628821, 0.560535, 0.670315, 0.647391]
  Kp [steer,rear] [10.2764, 0]; Kd [0.200364, 0]
  spawn roll -0.0387032 rad; rate -0.189371 rad/s
  gyro bias XYZ rad/s [-2.12844e-05, 3.17022e-05, 0.000155771]
  gravity bias YZ [0.00176665, 0.00100496]
E2 [20.82, 50.82) s:
  mass x [1.03942, 1.09104, 1.02408, 0.815617]
  CoM XYZ mm [4.00533, 2.65287, 1.77476]
  mu_s [0.693205, 0.787792, 0.559809, 0.794338]
  mu_d [0.693205, 0.617134, 0.559809, 0.526209]
  Kp [steer,rear] [10.527, 0]; Kd [0.210141, 0]
  spawn roll 0.00214958 rad; rate 0.066284 rad/s
  gyro bias XYZ rad/s [-8.82509e-05, -1.80504e-05, 6.60728e-05]
  gravity bias YZ [0.00286012, 0.00110588]
E3 [50.82, 60.00) s:
  mass x [1.01917, 0.803289, 1.02937, 0.877224]
  CoM XYZ mm [0.233695, 0.62497, -1.17226]
  mu_s [0.531594, 0.713129, 0.592046, 0.659445]
  mu_d [0.531594, 0.713129, 0.534947, 0.547619]
  Kp [steer,rear] [10.654, 0]; Kd [0.187429, 0]
  spawn roll -0.0647806 rad; rate -0.00877151 rad/s
  gyro bias XYZ rad/s [-0.000406793, 0.000384208, -2.96344e-05]
  gravity bias YZ [-0.00204948, -0.00215862]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 55; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.06971, 0.972924, 1.00779, 1.19138]
  CoM XYZ mm [-3.20054, 2.46581, -4.66681]
  mu_s [0.71155, 0.757579, 0.633024, 0.791209]
  mu_d [0.71155, 0.70921, 0.568172, 0.767936]
  Kp [steer,rear] [10.3741, 0]; Kd [0.186761, 0]
  spawn roll 0.02939 rad; rate -0.0619636 rad/s
  gyro bias XYZ rad/s [-0.000345929, 6.10222e-05, 0.000149293]
  gravity bias YZ [0.000140127, -0.000249329]
E2 [30.00, 30.64) s:
  mass x [0.950851, 1.15439, 1.06268, 1.03878]
  CoM XYZ mm [1.79913, 1.45826, 3.13366]
  mu_s [0.541939, 0.633024, 0.611239, 0.640016]
  mu_d [0.534359, 0.568172, 0.611239, 0.513141]
  Kp [steer,rear] [10.104, 0]; Kd [0.200072, 0]
  spawn roll -0.0578729 rad; rate -0.213384 rad/s
  gyro bias XYZ rad/s [-0.000286517, 0.000241971, 0.000139544]
  gravity bias YZ [-0.00335196, 0.003988]
E3 [30.64, 31.08) s:
  mass x [0.841558, 0.811055, 0.850415, 0.838444]
  CoM XYZ mm [0.964484, -2.95318, 3.32143]
  mu_s [0.723429, 0.704749, 0.640867, 0.541939]
  mu_d [0.723429, 0.541926, 0.556841, 0.534359]
  Kp [steer,rear] [9.83051, 0]; Kd [0.185102, 0]
  spawn roll -0.0238694 rad; rate 0.238069 rad/s
  gyro bias XYZ rad/s [0.000296954, 1.99077e-05, -5.1428e-05]
  gravity bias YZ [0.00185396, 0.00271763]
E4 [31.08, 31.64) s:
  mass x [0.977562, 0.939718, 1.10087, 0.859839]
  CoM XYZ mm [-0.713412, -2.41105, 3.34113]
  mu_s [0.669571, 0.611239, 0.700521, 0.631146]
  mu_d [0.669571, 0.611239, 0.641329, 0.631146]
  Kp [steer,rear] [10.5322, 0]; Kd [0.219664, 0]
  spawn roll -0.0492317 rad; rate 0.0183658 rad/s
  gyro bias XYZ rad/s [-0.000102469, -0.000191119, -0.000104485]
  gravity bias YZ [-0.00264815, -0.00138903]
E5 [31.64, 31.92) s:
  mass x [0.807522, 0.950731, 0.913443, 0.939251]
  CoM XYZ mm [3.64069, 3.87714, 0.785652]
  mu_s [0.640867, 0.723429, 0.704749, 0.538976]
  mu_d [0.556841, 0.723429, 0.541926, 0.538976]
  Kp [steer,rear] [9.35143, 0]; Kd [0.215904, 0]
  spawn roll -0.0752084 rad; rate 0.00748958 rad/s
  gyro bias XYZ rad/s [0.000333094, 1.15888e-05, -3.8052e-05]
  gravity bias YZ [-0.00305434, -0.000285825]
E6 [31.92, 32.28) s:
  mass x [0.915522, 1.19818, 1.06993, 0.841503]
  CoM XYZ mm [-3.59426, -3.46838, 2.35408]
  mu_s [0.71008, 0.733515, 0.611239, 0.748289]
  mu_d [0.59371, 0.558392, 0.611239, 0.689499]
  Kp [steer,rear] [9.17657, 0]; Kd [0.204401, 0]
  spawn roll 0.0669363 rad; rate 0.246136 rad/s
  gyro bias XYZ rad/s [2.15728e-05, -0.000658089, -0.000151426]
  gravity bias YZ [-0.000504504, -0.00157344]
E7 [32.28, 33.32) s:
  mass x [1.11837, 1.09435, 0.803739, 0.918133]
  CoM XYZ mm [-4.62211, 3.42452, -4.20362]
  mu_s [0.71008, 0.791209, 0.700521, 0.791209]
  mu_d [0.59371, 0.767936, 0.641329, 0.767936]
  Kp [steer,rear] [9.1785, 0]; Kd [0.192983, 0]
  spawn roll -0.0799881 rad; rate -0.0901051 rad/s
  gyro bias XYZ rad/s [-0.000159882, 2.21676e-05, 0.000189237]
  gravity bias YZ [0.000174635, -0.000361986]
E8 [33.32, 34.26) s:
  mass x [1.18228, 0.893188, 1.11623, 0.916708]
  CoM XYZ mm [3.98366, 0.741484, -3.22077]
  mu_s [0.559972, 0.538434, 0.749678, 0.732187]
  mu_d [0.502209, 0.538434, 0.741429, 0.631664]
  Kp [steer,rear] [9.21255, 0]; Kd [0.210853, 0]
  spawn roll -0.0273076 rad; rate -0.0692347 rad/s
  gyro bias XYZ rad/s [-6.51231e-05, 0.000370741, -0.000267846]
  gravity bias YZ [0.00228716, -0.00299861]
E9 [34.26, 35.00) s:
  mass x [1.12591, 0.900137, 0.954882, 0.933317]
  CoM XYZ mm [4.09999, -3.45071, -0.619425]
  mu_s [0.757579, 0.733515, 0.640016, 0.749678]
  mu_d [0.70921, 0.558392, 0.513141, 0.741429]
  Kp [steer,rear] [10.0242, 0]; Kd [0.20782, 0]
  spawn roll -0.0288042 rad; rate -0.0110942 rad/s
  gyro bias XYZ rad/s [-0.000205313, 0.000134744, 1.35886e-05]
  gravity bias YZ [0.00112404, 0.00168542]
E10 [35.00, 35.78) s:
  mass x [1.13111, 1.17678, 0.979955, 0.806021]
  CoM XYZ mm [1.04925, 2.58589, -1.10245]
  mu_s [0.568073, 0.791209, 0.69316, 0.640867]
  mu_d [0.568073, 0.767936, 0.69316, 0.556841]
  Kp [steer,rear] [9.52077, 0]; Kd [0.190062, 0]
  spawn roll 0.00935767 rad; rate -0.217517 rad/s
  gyro bias XYZ rad/s [0.000296809, -0.000175332, -0.000138796]
  gravity bias YZ [-0.00145292, 0.0010066]
E11 [35.78, 36.58) s:
  mass x [0.968944, 0.916349, 0.881028, 0.899013]
  CoM XYZ mm [-1.30586, -3.42872, -0.650544]
  mu_s [0.748289, 0.69316, 0.728342, 0.666375]
  mu_d [0.689499, 0.69316, 0.728342, 0.519145]
  Kp [steer,rear] [10.9664, 0]; Kd [0.215449, 0]
  spawn roll -0.00875553 rad; rate 0.0956034 rad/s
  gyro bias XYZ rad/s [-0.000132225, -1.92237e-05, -1.94566e-05]
  gravity bias YZ [0.000422119, 0.00267768]
E12 [36.58, 60.00) s:
  mass x [0.938605, 0.838997, 1.12104, 0.999597]
  CoM XYZ mm [-2.19752, -1.34488, -1.573]
  mu_s [0.700521, 0.723429, 0.631146, 0.669571]
  mu_d [0.641329, 0.723429, 0.631146, 0.669571]
  Kp [steer,rear] [9.02062, 0]; Kd [0.200193, 0]
  spawn roll -0.013736 rad; rate -0.118537 rad/s
  gyro bias XYZ rad/s [-0.000175155, -0.00023395, -5.02368e-05]
  gravity bias YZ [-0.00209707, 0.00430118]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 55; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.06971, 0.972924, 1.00779, 1.19138]
  CoM XYZ mm [-3.20054, 2.46581, -4.66681]
  mu_s [0.71155, 0.757579, 0.633024, 0.791209]
  mu_d [0.71155, 0.70921, 0.568172, 0.767936]
  Kp [steer,rear] [10.3741, 0]; Kd [0.186761, 0]
  spawn roll 0.02939 rad; rate -0.0619636 rad/s
  gyro bias XYZ rad/s [-0.000345929, 6.10222e-05, 0.000149293]
  gravity bias YZ [0.000140127, -0.000249329]
E2 [30.00, 60.00) s:
  mass x [0.950851, 1.15439, 1.06268, 1.03878]
  CoM XYZ mm [1.79913, 1.45826, 3.13366]
  mu_s [0.541939, 0.633024, 0.611239, 0.640016]
  mu_d [0.534359, 0.568172, 0.611239, 0.513141]
  Kp [steer,rear] [10.104, 0]; Kd [0.200072, 0]
  spawn roll -0.0578729 rad; rate -0.213384 rad/s
  gyro bias XYZ rad/s [-0.000286517, 0.000241971, 0.000139544]
  gravity bias YZ [-0.00335196, 0.003988]
Six significant digits shown; full precision: sampled_dr_values.json.
```